

Classroom Adaptive Sandbox for Tactile Learning and Engagement



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GitHub Repository: github.com/jm-Ferguson/CASTLE.

1. Abstract

1.1 Executive Summary

An interactive, sensor-driven tactile environment (potentially utilizing sand or similar physical mediums) designed as a responsive sensory learning tool for neurodivergent individuals. The objective is to build a working prototype and validate it through testing.

1.2 Core Idea

Create a physical sensory station that uses embedded sensors to detect user interaction and translate those physical movements into real-time digital feedback (such as lights, projections, or sounds) to aid in learning.

1.3 Key Concepts

Embedded systems, sensor integration (depth cameras, pressure, or motion sensors), real-time signal processing, human-computer interaction (HCI), and dynamic feedback mechanisms.

1.4 Problem Statement

Neurodivergent individuals can have different ways of processing information and maintaining attention, which can make traditional learning environments difficult to utilize. Most traditional learning methods rely on visual or auditory instructions. Sensory based learning provides another way to learn through physical interaction and multiple senses. However, there is a need for an interactive sensory learning tool that connects physical interaction with educational content and provides feedback. Recent literature supports the development of such tools; preliminary

findings suggest that “Tangible User Interfaces (TUIs) may enhance learning outcomes for children with ASD, warranting further investigation into their potential benefits” [1].

1.5 Proposed Solution

A hybrid hardware and software system that combines a physical tactile medium with sensors and real time feedback. The system will detect user interaction and translate it to corresponding responses through light, sound, haptics, or visual information.

2. Problem Description

2.1 Real World Problem

The modern educational system isn’t designed with neurodivergent students in mind, oftentimes neurodivergent students struggle in regular classrooms when it comes to teaching and sensory obstacles [2]. Another problem is that somatic senses, such as balance, rotation, and proprioception, are more pronounced in neurodivergent persons causing self-soothing tactics in an attempt to control the amount of stimulus they feel [3]. These tactics can be seen as distractions within a classroom, however, those faced with the difficulty of overactive stimulation often reportedly need to exhibit this behavior in order to stay focused and more adequately learn [3]. In addition to this, recent studies show that 20% of challenges faced by neurodivergent students are limited educator support along with another 20% challenges being inflexible assessments [4].

2.2 Why This Problem is Important

This problem is important because everyone deserves the best education possible. If using a tactile approach to learning with the assistance of modern technology can help these students learn, it should be worth pursuing.

2.3 The Engineering Problem

There are already discussions about trying to transition to technologically mediated models from traditional models for special education [5]. The main engineering problem is designing an Augmented Reality (A.R.) sandbox that is safe to use and is durable. To that extent, preventative measures must be taken to ensure sand does not harm any of the components of the system, as well as limiting access to the system's delicate hardware. The foundation of the structure must also be durable enough to withstand destructive behavior and continuous use.

2.4 The Relevance and Significance

Neurodivergent children, while just as ready to learn as their neurotypical peers, oftentimes need their environment to be adjusted. Providing neurodivergent children with a tactile means to learn a curriculum is one step closer to giving them the environment they need. While interactive sandboxes already exist like Topobox's AR sandbox, none of those sandboxes are focused on providing a medium for an expanded curriculum [6].

3. Background and Literature Review

3.1 Sensory Adaptive Learning Tool

Tancredi proposed a design that integrates sensory regulation with conceptual learning. By

correlating students' rocking movements to x, and y coordinates on a graph, Tancredi was able to demonstrate the mathematical concepts of sine waves to students. There are existing sensory accommodations like alternative seating and sensory tools that allow students to regulate stimulation while within a classroom. Existing systems often separate the act of sensory stimulation from the learning content of the classroom. Tancredi's Balance Graphing system addressed this problem by allowing children to generate projected mathematical graphs through adjustable rocking movements. The results of this study found that students could successfully use movement for operating the system as well as regulating sensory stimulation, participate in meaningful mathematical ideas, and communicate their own cognitive reasoning of the process. This provides justification for the Classroom Adaptive Sandbox for Tactile Learning and Engagement's (CASTLE) approach to a learning tool that combines the physical sensation of manipulating sand and real-time feedback for educational purposes. CASTLE extends this idea by providing a smooth and comforting tactile surface capable of hosting a multitude of academic activities [3].

3.2 Augmented Reality Sandboxes

Augmented reality (AR) sandboxes are not a new technology. They have been deployed in museums and schools across the United States. Products like Topobox have introduced the idea of an interactive sandbox for the purpose of a learning aid. Topobox combines a distance camera, projector, and computing unit to provide a real-time interactive sandbox experience. [6] This gives justification that CASTLE can take a similar approach using applicable hardware and software. These tools, however, focus primarily on geography through interactive topological learning tools. CASTLE seeks to expand this idea into other areas of learning, while focusing

primarily on students who struggle with traditional learning means. Using this medium to bolster sensory-adaptive embodied learning as well as student assisted learning. [6]

4. Proposed System Features

1. Interactive/Reactive Sand-based Surface

- a. Multiple sensors (depth camera, camera, ultrasonic receivers/transmitters) which would allow the system to know how and where a hand is positioned within the sandbox, depth of the sand within the sandbox, the amount of sand within the sandbox, and objects within the sandbox.
- b. The device would react to inputs in multiple ways. The first being a projection onto the sand which would provide visual feedback to the students using the sandbox. Other outputs of the system would be auditory soft chimes, or haptic feedback to communicate a correct answer, or respond to a student, and provide other stimulus to the student.

2. Durable, Safe, and Mobile Body

- a. Wooden base that can hold up the sandbox and support a significant amount of weight.
- b. Fully enclosed bottom to hide and lock away fragile hardware components
- c. Wheels attached to the bottom of the frame to make the system mobile and maneuverable
- d. Electric locking lid on top of the box for storage and prevention of unintended access to sandbox when not in use

3. Completely Modular Design.

- a. Designed software with adaptations and total overhaul in mind. The goal would be to create an easy way to add ‘learning activities’ into the device for future growth of the curriculum used within the sandbox.

5. Timeline

Week	Task
1 - 3	Finalize Team, Idea, and Proposal
4	Submit & Present Proposal
5	Draft of Software Requirements Specifications (SRS)
6	Final Software Requirements Specifications (SRS)
7 - 8	Start High Level Design (HLD)
9	Submit High Level Design (HLD) Draft
10 - 13	High Level Design Report & Supported Meetings
14 -16	Final Report & Presentation

Table 5.1; Capstone Timeline - Semester One

6. Budget & Resources

Item	Description	Price
Wood	Foundational material, used in the general construction of our prototype. (<i>Wood provided by a group member</i>)	\$0
Sand	‘Washed’ play-sand.	\$53.74
Screws	Used for fastening wood pieces together.	\$11.74
Velcro Wire Strap	Used for wire-management.	\$6.41
Projector	‘Screen’ Projection onto the top layer of sand, allowing interactive feedback.	\$113.39
Camera	Top-down facing camera, to allow for hand tracking, etc.	\$26.74
Depth Camera	Top-down facing depth camera, to allow the system to understand the topology of the sand within the sandbox.	\$85.59
Computer Hardware	Generic computer parts consisting of a SSD, Graphics Card, CPU, RAM, and a Power Supply.	\$255.73
Arduino Uno	Arduino Uno w/ ATMEGA 328p, used for control of Ultrasonic transmitters/receivers.	\$29.54
Wire	Wire used to connect the Ultrasonic System together.	\$17.11
Ultrasonic Kit	Kit including transmitters and receivers.	\$10.58
Electronic Latch	Electronically controlled latch to create a locking mechanism to ensure safety when the device isn't being used.	\$11.03
HDMI Cables	Display cables used for connecting devices together.	\$10.69
Power Strip	Surge protector, plus a powerstrip to allow for multiple devices to be powered at the same time.	\$9.99
Usb Extender	Usb extender to ensure camera cables can reach our computer hardware.	\$5.64
Extension Cord	Extension cable to ensure all devices can be powered despite cable length.	\$7.48

Table 6.1; Tentative Project Budget

When budgeting for this project, we attempted to stay within the scope of creating a fundamentally working prototype which would allow us to cut costs from buying premium materials and focus on a working idea. Below is a table illustrating our current expected budget, and each item within that budget. Clearly separated by type of items like **Materials**, **Fasteners**, and **Electrical Components** (see table 6.1). After finalizing our budget, we determined that approximately \$655.40 would be required as a baseline cost in developing a working prototype. This estimate accounts for the materials and components we think are necessary to construct and implement the prototype.

7. References

- [1] M. De Luca, *et al.*, "Using Tangible User Interfaces (TUIs): Preliminary Evidence on Memory and Comprehension Skills in Children with Autism Spectrum Disorder," *Behavioral Sciences*, vol. 15, no. 3, p. 267, 2025, doi: 10.3390/bs15030267.
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- [3] S. Tancredi, "Sensory-adaptive embodied design: Integrating movement for sensory regulation and conceptual learning," *Multimodality & Society*, University of Wisconsin–Madison, 2026.
- [4] T. Fernando, "Embracing Neurodiversity in Education: A Review of Inclusive Practices, Policies, and Pedagogies," *SchoRes Journal of Education Research*, vol. 1, no. 2, pp. 75–81, 2024. Available: <https://schores.org/journals/sjer/article/view/25>
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- [6] TopoBox, "Augmented reality sandbox, AR sandbox," *TopoBox*. [Online]. Available <https://www.topobox.co>. [Accessed: Sept 1. 2026]